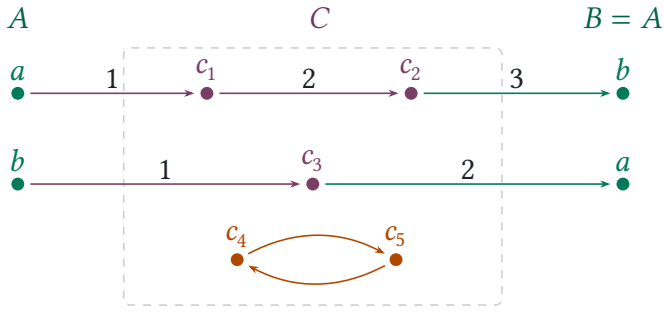
(a) One application of f

$$A \sqcup C \xrightarrow{f} B \sqcup C$$

$$a \mapsto c_1$$
$$b \mapsto c_3$$
$$c_1 \mapsto c_2$$
$$c_2 \mapsto b$$
$$c_3 \mapsto a$$
$$c_4 \mapsto c_5$$
$$c_5 \mapsto c_4$$

(b) Feed only C back



$$\underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}}_{x_0} + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{x_1} + \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{x_2} = \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\mathrm{Tr}^C f}$$